■ Torch Experiment

A Hands-On Lesson on Hardware, Circuits & Electricity

Grade Level:	Grade 2
Subject:	Science / STEM (Hardware & Electricity)
Duration:	45–60 minutes
Lesson Topic:	Building and understanding a simple torch (flashlight) circuit

■ Learning Objectives

By the end of this lesson, students will be able to:

- Define **hardware** as anything tangible — tools, machines, and gadgets we can touch and use.
- Identify a **torch** as a handheld light tool that runs on batteries.
- Name the four main parts of a torch: **battery, switch, bulb/LED, and torch body**.
- Explain how electricity flows from the battery through wires to the bulb in a **circuit**.
- Build a simple torch model with adult supervision, following safety guidelines.
- Describe real-world uses of torches (camping, power cuts) and a bit of their history.

■ Materials Needed

Gather these items *before* the lesson. An adult should supervise collection and use of sharp tools.

- Torch body (cardboard tube or plastic tube)
- 1–2 batteries
- A small bulb or LED
- Connecting wires
- A switch
- Tape and scissors (adult-assisted)

■ Key Vocabulary

Term	Kid-Friendly Definition
Hardware	Anything you can touch and hold — tools, machines, and gadgets.
Torch	A small handheld light that runs on batteries.

Battery	Stores energy and powers the torch.
Switch	Turns the torch on and off by controlling the flow of electricity.
Bulb / LED	Lights up when electricity flows through it.
Torch Body	Holds all the parts together safely.
Circuit	A loop that electricity flows through, from battery to bulb and back.

■ Lesson Flow

1. Introduction: What is Hardware? (5 minutes)

Begin by asking students: *'What is something you can touch and hold?'* Explain that hardware is anything tangible — tools, machines, and gadgets like screwdrivers, batteries, and torches. Discuss how hardware makes daily life easier and brighter.

2. What is a Torch? (5 minutes)

Introduce the torch as a small handheld light that runs on batteries and has a bulb or LED that glows brightly. Ask students when they have used a torch (camping, power outages, looking under furniture). Emphasize: a torch helps us see in the dark!

3. Parts of a Torch (10 minutes)

Walk through the four main parts and their jobs:

- **Battery** — stores energy and powers the torch
- **Switch** — turns the torch on and off
- **Bulb or LED** — lights up when electricity flows through it
- **Torch Body** — holds all the parts together safely

4. How Does a Torch Work? (5 minutes)

Explain that pressing the switch lets electricity travel from the battery through the wires to light up the bulb. Energy flows in a loop called a **circuit**. Key idea: *No electricity = no light. Close the circuit = bright light!*

5. Hands-On Activity: Build Your Own Torch (20–25 minutes)

Guide students through building a simple torch model. **An adult must supervise every step, especially anything involving scissors or sharp tools.**

Step 1 – Prepare the Torch Body

Take a cardboard or plastic tube. With adult help, carefully make a small hole at one end where the bulb will go. The hole should be just the right size — not too big, not too small.

Step 2 – Connect the Battery and Bulb

Use wires to connect the battery to the bulb. Twist the wire ends tightly to the battery terminals and bulb holder so electricity can flow properly. Check connections before testing.

Step 3 – Add the Switch

Connect the switch between the battery and the bulb using wires. Make sure the switch clicks when pressed, and ask an adult to fix it firmly in place. Remind students: switch ON means light, switch OFF means dark.

Step 4 – Test Your Torch!

Press the switch and watch the bulb light up. If it doesn't work, check the wires and connections, then try again.

6. What Happens Inside the Torch? (5 minutes)

Recap the electricity flow: **Battery** → **Wires** → **Bulb = Light!** Electricity moves from the battery through the wires to the bulb, lighting it up. This energy travels in a loop called a circuit.

■■ Safety Tips

- Always ask an adult for help with wires and scissors.
- Don't use old or damaged batteries — they can leak!
- Keep your workspace clean and tidy at all times.

■ Fun Facts About Torches

- The first torches used candles for light!
- Modern torches use LEDs that last a very long time.
- Torches help us during camping and power cuts!

■ Recap / Discussion Questions

Use these questions to check understanding at the end of the lesson:

1. What are the main parts of a torch?
2. What makes the bulb light up?
3. Why do we need a switch?
4. What should you do before starting the experiment?

■ Extension Activity / Next Steps

Encourage students to try building other simple circuits at home with the help of a parent or teacher — for example, a circuit with a buzzer instead of a bulb, or one with multiple switches. Ask students to keep their finished torches as a reminder of how electricity powers everyday hardware.

■ Great job, Superstar Scientists! Keep exploring and stay curious!